7.8.1 Computing the Correction Functions $f_1(q,p)$

At this point we may plow ahead and compute the PDE for each corresponding scheme, but a more efficient route is offered by working with the structure of the inhomogeneity in (7.21). We may make use of the identity

$$\left([\mathcal{L}_{\mathrm{O}}^{\dagger}, \mathcal{L}_{\mathrm{A}}^{\dagger}] + [\mathcal{L}_{\mathrm{O}}^{\dagger}, \mathcal{L}_{\mathrm{B}}^{\dagger}]\right) \rho_\beta = \left([\mathcal{L}_{\mathrm{O}}^{\dagger}, \mathcal{L}_{\mathrm{A}}^{\dagger} + \mathcal{L}_{\mathrm{B}}^{\dagger}]\right) \rho_\beta = 0,$$

as of course $\mathcal{L}_H^{\dagger} = \mathcal{L}_{\mathrm{A}}^{\dagger} + \mathcal{L}_{\mathrm{B}}^{\dagger}$ is the Liouvillian of our Hamiltonian system, which preserves any function of the energy. Thus, by the skew-symmetry of the commutator ($[P,Q] = -[Q,P]$) we can relate four of the six methods as

$$\mathcal{L}_{1,[\![\mathrm{ABO}]\!]}^{\dagger}\rho_\beta = \mathcal{L}_{1,[\![\mathrm{OAB}]\!]}^{\dagger}\rho_\beta = -\mathcal{L}_{1,[\![\mathrm{OBA}]\!]}^{\dagger}\rho_\beta = -\mathcal{L}_{1,[\![\mathrm{BAO}]\!]}^{\dagger}\rho_\beta = -\frac{1}{2}[\mathcal{L}_{\mathrm{A}}^{\dagger}, \mathcal{L}_{\mathrm{B}}^{\dagger}]\rho_\beta, \tag{7.22}$$

by virtue of (7.20). Hence the leading order correction functions of the schemes are related through

$$f_{1,[\![\mathrm{ABO}]\!]} = f_{1,[\![\mathrm{OAB}]\!]} = -f_{1,[\![\mathrm{OBA}]\!]} = -f_{1,[\![\mathrm{BAO}]\!]}. \tag{7.23}$$

The remaining two permutations of the possible $[\![\mathrm{XYZ}]\!]$ schemes have a similar relationship, with

$$\mathcal{L}_{1,[\![\mathrm{AOB}]\!]}^{\dagger}\rho_\beta = -\mathcal{L}_{1,[\![\mathrm{BOA}]\!]}^{\dagger}\rho_\beta = -\frac{1}{2}[\mathcal{L}_{\mathrm{A}}^{\dagger} - \mathcal{L}_{\mathrm{B}}^{\dagger}, \mathcal{L}_{\mathrm{O}}^{\dagger}]\rho_\beta + \mathcal{L}_{1,[\![\mathrm{ABO}]\!]}^{\dagger}\rho_\beta. \tag{7.24}$$

This reduces our workload as we can find the leading order behavior for all six methods by computing the solution to two PDEs. This is not necessarily trivial, but it turns out that the structure of the equations is simple enough that we can find exact solutions. We work now to compute the right-hand side of the governing PDE given in (7.21).

In order to solve (7.21) for the $[\![\mathrm{ABO}]\!]$ scheme, we need to compute the right hand side of (7.22). First, note that if we define the function $g(\boldsymbol{q},\boldsymbol{p})$ as

$$g(\boldsymbol{q},\boldsymbol{p}) = \beta \boldsymbol{p}^T \boldsymbol{M}^{-1} \nabla U(\boldsymbol{q}),$$

then through simple computation we have

$$\mathcal{L}_{\mathrm{A}}^{\dagger}\rho_\beta = -\mathcal{L}_{\mathrm{B}}^{\dagger}\rho_\beta = g(\boldsymbol{q},\boldsymbol{p})\rho_\beta.$$

This is particularly useful as

$$\begin{aligned}[\mathcal{L}_{\mathrm{A}}^{\dagger}, \mathcal{L}_{\mathrm{B}}^{\dagger}]\rho_\beta &= \left(\mathcal{L}_{\mathrm{A}}^{\dagger}\mathcal{L}_{\mathrm{B}}^{\dagger} - \mathcal{L}_{\mathrm{B}}^{\dagger}\mathcal{L}_{\mathrm{A}}^{\dagger}\right)\rho_\beta = -\left(\mathcal{L}_{\mathrm{A}}^{\dagger} + \mathcal{L}_{\mathrm{B}}^{\dagger}\right)\left(g(\boldsymbol{q},\boldsymbol{p})\rho_\beta\right) \\ &= \left(\mathcal{L}_{\mathrm{O}}^{\dagger} - \mathcal{L}_{\mathrm{LD}}^{\dagger}\right)\left(g(\boldsymbol{q},\boldsymbol{p})\rho_\beta\right).\end{aligned}$$

Now notice that for a function $\psi(\boldsymbol{q})$ purely of position, we have

$$\mathcal{L}_{\mathrm{LD}}^{\dagger}\left(\psi(\boldsymbol{q})\rho_\beta\right) = -\left(\boldsymbol{p}^T\boldsymbol{M}^{-1}\nabla\psi(\boldsymbol{q})\right)\rho_\beta(\boldsymbol{q},\boldsymbol{p}),$$

and hence we can compute

$$\begin{aligned}\mathcal{L}_{\mathrm{O}}^{\dagger}\left(g(\boldsymbol{q},\boldsymbol{p})\rho_\beta\right) &= \gamma\rho_\beta\left(-p\cdot\nabla_p g(\boldsymbol{q},\boldsymbol{p}) + \beta^{-1}\nabla_p\cdot\boldsymbol{M}\nabla_p g(\boldsymbol{q},\boldsymbol{p})\right), \\ &= -\gamma g(\boldsymbol{q},\boldsymbol{p})\rho_\beta, \\ &= \gamma\beta\mathcal{L}_{\mathrm{LD}}^{\dagger}\left(U(\boldsymbol{q})\rho_\beta\right).\end{aligned}$$

This means that

$$\mathcal{L}_{1,[\![\mathrm{ABO}]\!]}^{\dagger}\rho_\beta = -\frac{1}{2}[\mathcal{L}_{\mathrm{A}}^{\dagger}, \mathcal{L}_{\mathrm{B}}^{\dagger}]\rho_\beta = -\frac{1}{2}\mathcal{L}_{\mathrm{LD}}^{\dagger}\left(\left[\gamma\beta U(\boldsymbol{q}) - g(\boldsymbol{q},\boldsymbol{p})\right]\rho_\beta\right).$$

Hence, we can combine this equation with (7.21) to yield

$$\mathcal{L}_{\mathrm{LD}}^{\dagger}f_{1,[\![\mathrm{ABO}]\!]}\rho_\beta = \frac{1}{2}\mathcal{L}_{\mathrm{LD}}^{\dagger}\left(\left[\gamma\beta U(\boldsymbol{q}) - g(\boldsymbol{q},\boldsymbol{p})\right]\rho_\beta\right),$$

with the general solution

$$f_{1,[\![\mathrm{ABO}]\!]} = \frac{\beta}{2}\left(\gamma U(\boldsymbol{q}) - \boldsymbol{p}^T\boldsymbol{M}^{-1}\nabla U(\boldsymbol{q}) - c\right),$$

where $c = \gamma\langle U(\boldsymbol{q})\rangle$ is a constant, chosen to ensure that the average of the correction function is zero. This solution gives the leading order correction for the three related schemes in (7.23).

For the final two schemes, we can compute the correction function through the use of (7.24). Our previous work has computed the second term on the right hand side already, so we are left with the initial term. However, its expression reduces quite easily, as

$$\begin{aligned}[\mathcal{L}_{\mathrm{A}}^{\dagger} - \mathcal{L}_{\mathrm{B}}^{\dagger}, \mathcal{L}_{\mathrm{O}}^{\dagger}]\rho_\beta &= -\mathcal{L}_{\mathrm{O}}^{\dagger}\left(\mathcal{L}_{\mathrm{A}}^{\dagger} - \mathcal{L}_{\mathrm{B}}^{\dagger}\right)\rho_\beta, \\ &= -2\mathcal{L}_{\mathrm{O}}^{\dagger}\left(g(\boldsymbol{q},\boldsymbol{p})\rho_\beta\right), \\ &= -2\gamma\beta\mathcal{L}_{\mathrm{LD}}^{\dagger}\left(U(\boldsymbol{q})\rho_\beta\right).\end{aligned}$$